

Surrogate Variable Analysis (SVA)

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Introduction

Surrogate variable analysis (SVA) estimates latent factors that capture unmodelled heterogeneity in high-dimensional expression data – unknown batch effects, sample-processing artefacts, or unmeasured biology correlated with technical variables. Including surrogate variables as covariates in downstream regression restores power and controls false positives where ComBat cannot apply (because the batch variable is itself unknown).

Prerequisites

Linear models on expression data, batch effects, ComBat.

Theory

SVA decomposes the residual expression matrix (after regressing out the primary model) into latent factors using a bootstrap-based singular value decomposition. The number of surrogate variables k is chosen by a permutation test (`num.sv`). Each surrogate is then added as a covariate in the downstream limma/edgeR model.

For RNA-seq count data, use `svaseq`, which applies SVA on a log-transformed scale suitable for negative-binomial outcomes.

Assumptions

The primary variable of interest is modelled correctly; surrogate variables are orthogonal to it; residual variation reflects technical or unmodelled factors.

R Implementation

```
library(sva); library(limma)

set.seed(2026)
n_genes <- 3000; n_samples <- 20
group <- factor(rep(c("ctrl", "trt"), each = 10))
batch <- factor(rep(c("A", "B"), times = 10)) # hidden

y <- matrix(rnorm(n_genes * n_samples), n_genes, n_samples)
y <- y + matrix(rep(as.numeric(batch) * 0.6, each = n_genes),
                 n_genes, n_samples)

mod <- model.matrix(~ group)
mod0 <- model.matrix(~ 1, data = data.frame(group))

n_sv <- num.sv(y, mod, method = "be")
svobj <- sva(y, mod, mod0, n.sv = n_sv)
```

```
mod_sv <- cbind(mod, svobj$sv)
fit <- lmFit(y, mod_sv); fit <- eBayes(fit)
topTable(fit, coef = 2, n = 5)
```

Output & Results

svobj\$sv contains $n_s v$ surrogate variables per sample. Including them typically recovers differential expression that batch confounding otherwise hides.

Interpretation

“We estimated k surrogate variables using SVA and included them in the limma model, adjusting for unmodelled heterogeneity without needing to specify a batch variable.”

Practical Tips

- Use `svaseq` for RNA-seq count data.
- The `num.sv(..., method = "be")` Buja-Eyuboglu estimator is conservative; "`leek`" is more aggressive.
- Surrogates should not correlate with the variable of interest; `sva` orthogonalises them by construction.
- Always inspect what surrogates correlate with (library size, processing date) for interpretation.
- If the batch is known, prefer ComBat-seq or direct covariate inclusion; SVA is for hidden factors.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren't.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- RNA-seq with DESeq2 / edgeR
- Enrichment analysis (GSEA / ORA)
- scRNA-seq with Seurat